# Giovanni Capuano

Senior Software Engineer

Based in: **London**

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## About me



**Trust, vision, meaning**. With these values as my guiding principles, I'm motivated by a desire to **collaborate** closely with innovators to bring their vision to life. At Honeypot we helped families living in difficult parts of the world make their dreams come true, and at Simply Business, we took care of all the legwork so that our customers could focus on what really matters, safely.

I aim to bring more than just technical expertise to the table. I adopt an **holistic** approach to application development focusing on the **user experience, product workflows, and the overall business impact**. Additionally, I want to help teams to find the practices that work best for them allowing to **perfection development experience for delivering quality work** that is easy to **maintain and evolve**.

I am passionate in designing and developing **small, fast, language-agnostic, disposable, well tested and carefully documented** pieces of softwares and architectures, which behaviors and interactions are easily **traced, monitored and observable**. Ecosystems engineers are glad and incentivised to work with and extend, feel **ownership** of and are keen to **experiment and innovate**.

**Ease of change, scalability, disaster recovery and keeping of decision records** are topics I value to stress early on.

Professionally, in the latest couple of years I have worked extensively with tools and technologies such as

- **Docker Terraform GitHub Actions** and **Semaphore CI**
- **Ruby on Rails Grape Kotlin/Spring Boot** and **NodeJS**
- **TypeScript** and **React** (design systems, Storybook, feature flags and A/B experiments)
- **PostgreSQL** and **MongoDB** (change streams, Atlas)
- **Kafka Kafka Connect** (MongoDB and PostgreSQL source connectors) and **Redis**
- **AWS S3 SQS** and **ECS**

- **Pacts** and **OpenAPI**
- **CDC/Event schemas** and the **transactional outbox** pattern
- **Stripe** and **TokenEx** payment integrations
- **Amplitude** experiments, **Snowplow** and **GTM** analytics
- **OpenTelemetry NewRelic Backstage**
- **RBS/Steep** gradual typing for Ruby
- **Solidity** and **Smart contracts**

I started my path as an engineer back in 2012 when I wanted to build applications for supporting my friends and me building tactics and strategies for competitive Pokémon battles. That also led me to approach web development first with the a classic `LAMP` stack and then then with `Ruby` (`Sinatra`, `Hanami` and `Ruby on Rails`), `Kotlin` and the `Rust` programming language, which I started studying since its beta stage. I also enjoy building web-components with both modern `Vanilla JS` and `React`.

I also grew curiosity towards Arduino, Raspberry PI, Android and so on, interests that gave me reasons to look into languages such as `Swift` and practice a bit more `Java` and `C`.

In 2015 I accepted to join a digital studio company to meet a bunch of highly skilled professionals which gave me the chance to further **grow** as human being and engineer, and at the same time having the opportunity to **shape products** that had a real **impact** on people's life

From there I joined Honeypot, a startup that was in the process of going live, in which I contributed as second in-house engineer to the foundation of the culture and tech infrastructure. During that period I also contributed to the organization of the first edition of RustFest (https://rustfest.global) which had place in Berlin on 2016.

Currently, at Simply Business my team mates (a brilliant, cohesive and passionate bunch!) and I have gone from building the **event-streaming backbone** of the company to rebuilding the **customer-facing sales journey and checkout**, while constantly **expanding and sharing** our knowledge, with frequent and efficient **feedback and retrospectives** - and still I enjoy writing code as hobbist, in the attempt to create something **meaningful** for me and others, or just having fun by doing it.

# Work experience

## Simply Business

**Senior Software Engineer**

October, 2019 - Present

*simplybusiness.com*

Acquired by Travelers [NYSE: TRV] in 2017 and based in London and Boston, Simply Business is the **leading online business insurance brokerage**: it protects over one million small businesses across thousands of trades in the **UK and the US**, matching each of them with the cover suited to its risks and priorities.

I joined in 2019 as a Senior Software Engineer in the team that owns the company's **data streaming backbone** and the back-office for **policy renewals and cancellations**; since late 2025 I have been rebuilding the **sales journey and checkout** in React and TypeScript. Rather than list everything, here is what the work has taught me to do well, with one example each.

**Taking a problem from ambiguity to production.** I am comfortable with the part before the code: discovery, spikes, writing the architecture decision record, deciding what not to build. For Record Verification, the service that proves the company's data lake matches production, I ran the discovery and the spikes, wrote the ADR and led the build to a finance audit deadline, then stayed for the aftercare until the alerts could be trusted.

**Designing systems that stay correct when things fail.** Most of my architecture work has been about consistency across boundaries. With my team I took the Kafka Outbox from a dual-write problem to the company standard for change data capture and domain events, over MongoDB and later PostgreSQL, with freshness SLOs, dead-letter runbooks and multi-region disaster recovery gamedays. The same instinct shows up in smaller places, such as idempotency keys and cart digests in the checkout.

**Making technical choices and bringing people along.** I will argue for an unconventional choice when it is the right one, and then make sure the team is not left behind: TypeScript for Record Verification in a Ruby organisation, SAML/Okta single sign-on replacing LDAP, the first Bootstrap upgrade in twelve years on the renewals application. Each one ended with the team able and happy to work on it, which I consider part of the deliverable.

**Leading work across teams.** I led the TokenEx payment migration to the new React architecture: built the first version and the patterns, then onboarded engineers from two teams so that we could finish it together. The outbox adoption worked the same way, migrating some fifty event types with the teams that owned them, and the rating logs extraction meant running the design conversations with the pricing, audit and rating teams.

**Full-stack product engineering.** I am at home on both sides of a typed contract: Rails page services and builders on one side, React components with their stories and tests on the other, with payments (Stripe, TokenEx), experiments (Amplitude) and analytics (Snowplow, GTM) in between. The checkout, coverage selection and confirmation pages of the sales journey are where this currently lands.

**Operating what we build.** Observability, alerting, runbooks and documentation are part of my definition of done, not a follow-up. I keep every service I touch documented on the internal developer portal, write and review incident reviews, and take my turn on incidents and alerts. I have kept the renewals application current for six years, from Rails 5.2 to 8.

**Making the team work well.** I pair and explain a lot, review quickly so that nobody waits on me, and facilitate retrospectives, risk storming and similar sessions, each designed around the context and the goals the project has at that moment, trying new techniques and asking for feedback to refine them. When ownership moves I hand it over with written guides and code walkthroughs. I interview engineers applying to join us.

**Working with AI.** I use AI coding agents every day and maintain the team's agent skills and guides for the component architecture, so that generated code fits the codebase and its conventions. Code is getting cheap; the judgement about what to build, how it fails and how the team learns is where I spend my effort.

**Selected work**

- 2025 - present: sales journey and checkout migration to React / TypeScript (coverage selection, confirmation pages, Combined Payment Page, Stripe and TokenEx, multi-product carts).
- 2025: Record Verification, a Node.js / TypeScript service proving the data lake matches production (MongoDB change streams, S3, Snowflake).
- 2024 - 2025: rating logs and insurer upsells moved into the rating service (OpenAPI, Pact, CDC in transactions).
- 2022 - 2025: Kafka Outbox, the company-wide CDC and domain events platform (MongoDB and PostgreSQL, Kafka Connect, Kotlin transformer, New Relic alerting, disaster recovery).
- 2019 - present: Continuous Cover, the UK and US renewals and cancellations back-office (US workflow, SAML/Okta, Bootstrap 5, Rails 8).
- 2020 - 2021: quoteset persistence moved from MongoDB to S3 behind a Mongoid-compatible driver, with no changes for callers.

# Honeypot

**Software Engineer**

November, 2019 - May, 2018

*honeypot.io*

Honeypot is Europe's developer-focused job platform that let companies do the first move, using a powerful set of **search engine filters and keywords** and also **receving tailored emails** showing talents that are most relevant for them. Recruiters are also able to connect to their account **ATS** such as *Greenhouse* or *Personio* to avoid any additional manual work.

I joined the company as **second in-house full-time engineer** and during the 3 years and half (then it got acquired by the German LinkedIn competitor: XING/NewWork) I have contributed to enstablish a **developer-focused culture, shaped a solid and mantainable infrastructure, sane and safe workflows** and at every opportunity shared knowledge for providing a **clear understanding** of the software development processes among all the non-tech colleagues.

- I have done **daily round of interviews** to developers applying to join the organization and worked with my colleagues on establishing a sound internal hiring process.
- Supported the **Business Intelligence** team with the creation of a foundation of databases and workflows to allow them applying their algorithms and analysis on sandboxed, reliable data.
- Built an open source search engine allowing the search for talents becoming dozen of times faster, with the service being resilient and rock solid. I started working on it when Rust was not yet v1.0 and many features and libraries were not stable or fully-functioning yet -

thus a lot of resources went to contributing to other projects (including the Rust buildpack for Heroku), to Rollbar and CircleCI and to the whole workflow to deploy and orchestrate all the different stages. Initial challenges aside, Rust proved to be a solid choice for such a kind of a project.
- Technologies I have been working with include: Ruby on Rails, Rust, Elixir, Ember.JS, Redis, Heroku, CircleCI and TravisCI, AWS Cloudfront and S3, Grape (RESTful APIs based on JWT and TOTP authentication), Postmark and Mailchimp, Greenhouse and Personio, MindMatch AI, TalkJS...

- https://github.com/honeypotio/reina – a CLI tool and GitHub bot for solving our need of orchestrating the deployments of the feature stagings of our applications on Heroku
- https://github.com/honeypotio/searchspot – ElasticSearch-backed client/server written in Rust used by recruiters to search the talents they are looking for
- https://github.com/benashford/rs-es – Rust client for the ElasticSearch REST API I contributed to in order to make Searchspot
- https://github.com/RoxasShadow/rollbar-rs/ – library to track and report errors, exceptions and messages from Searchspot application to Rollbar
- Blog posts such as https://blog.honeypot.io/errors-and-exceptions-in-rust

## Moze

**Software Engineer**

April, 2014 - October, 2015

*mozestudio.com*

Moze is an Italian studio of designers and developers which designs and builds digital products for entrepreneurs and forward-looking companies to shape or enhance digital products and services through a pragmatic design process and agile, incremental development.

I have been the first in-house full-time engineer after years the company relied mostly on contractors. I was close to every side of the projects, from the **front-end to the back-end to the dev-ops**.

- Among the other projects, we built Bugree: similarly to Proton Mail and MEGA, it was the first portal to make communications happens under NDA, and the first portal with **client-side encryption bounded to a dynamic key**, therefore messages never left any electronic footprint. There were more features like scrolling text, torch mode, limited openings and watermarks. Both messages and attachments were secured by end-to-end encryption and notifications were **real-time** thanks to a websockets-based NodeJS application we made. Its frontend was built with ReactJS and AngularJS and we implemented multiple payment processors including PayPal and Stripe.
- We also built GODO Shopping (for which we also made and open sourced zanox-rb, as we needed a good library to interact with the APIs of the popular affiliate marketing network service), which was a marketplace that with every purchase was granted a percentage in cashback to the buyer.

- Tourizan was another good project: like AirBnB but for enjoying unique experiences and places in Italy.

# Bio

Born near **Naples, Italy**. Graduated in scientific high school and attended **Computer Engineering** at the University of Naples "Federico II" before leaving my hometown to accept a role at Moze, in **Milan**.

Italian, Neapolitan, English and some Spanish aside, I have been studying **Japanese** and joined an intensive course in a language school in Kyoto, in the meantime I was living with a local family thanks to the homestay program.

I love travelling and discovering new cuisines of all world but also all the regional cuisines of Italy. I love to prepare food and experiment new dishes everyday in my kitchen (and share the results on my Instagram!).

Being on holiday means for me means hiking mountains with friends and go discovering new pizza places or enjoying the seasonal menus of my favorites!

I'm a also passioned videogamer since I have memory of: always thirst of superb game and level designs, great soundtracks and meaningful stories.

*Giovanni Capuano - hey@giovannicapuano.net - References on request*